

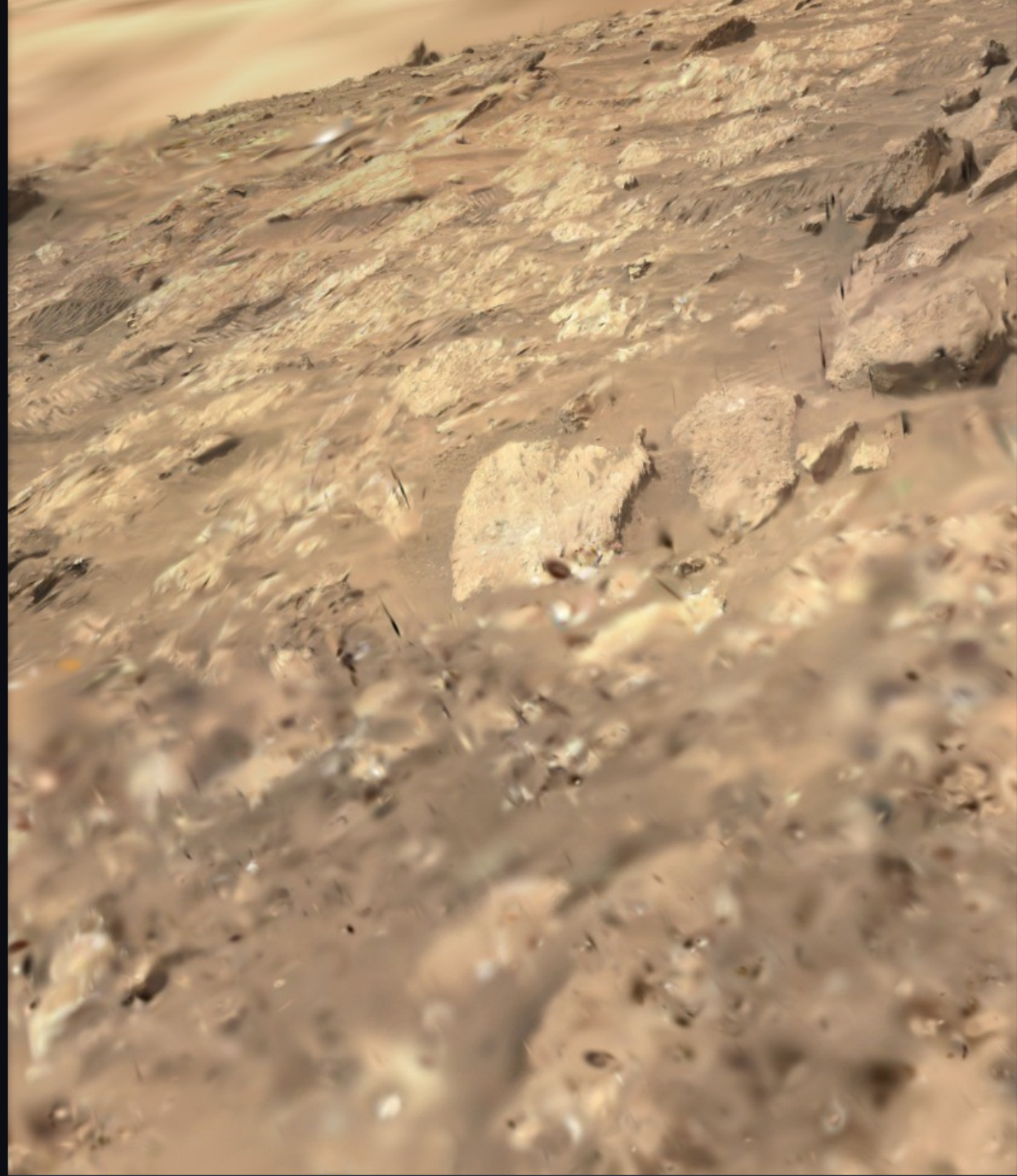
GAUSSIAN SPLATTING

From rover photos to a walkable piece of Mars

241 Perseverance photos of Cheyava Falls became 972,047 Gaussians, in real metres, without ever guessing where a camera was.

Right: our splat in the viewer, from rover height.

Planetary Scene Studio · MHacks



STEP 1

Start from real photos, from three parking spots

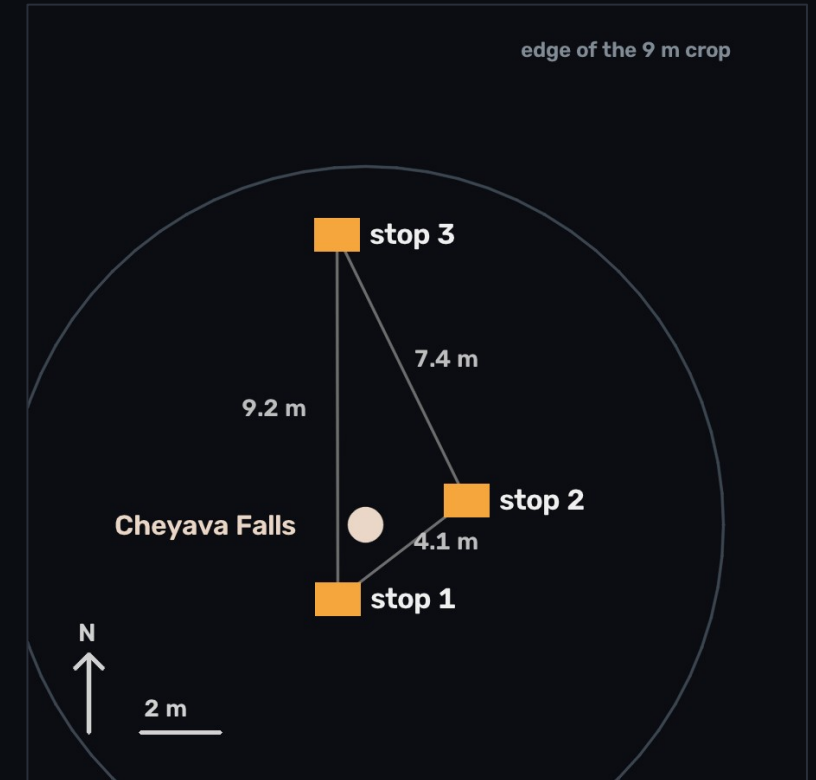
A splat needs the same ground seen from different places. One stop only gives rotation; three give depth.



Mastcam-Z: a rock and the rover's wheel tracks



Mastcam-Z: where the sample was drilled



Where the rover stood, to scale

241 photos from NASA's public Mars 2020 feed, sols 1195 to 1210

2 cameras Navcam (wide stereo pair) and Mastcam-Z (zoom, colour)

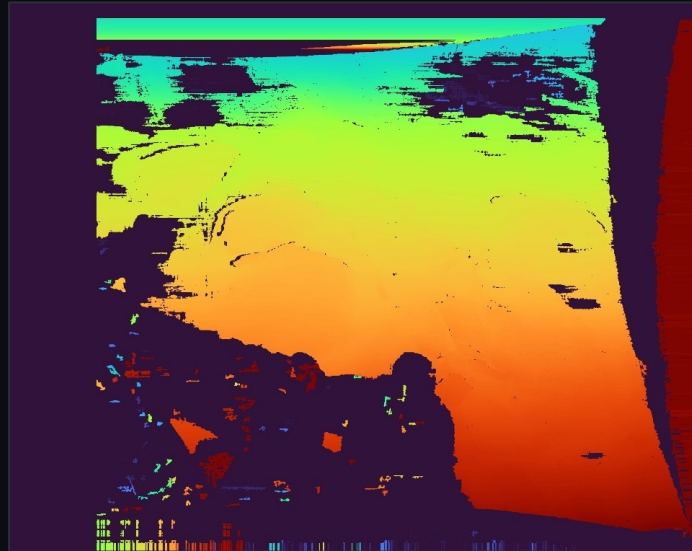
STEP 2

Get the geometry from NASA, not from guesswork

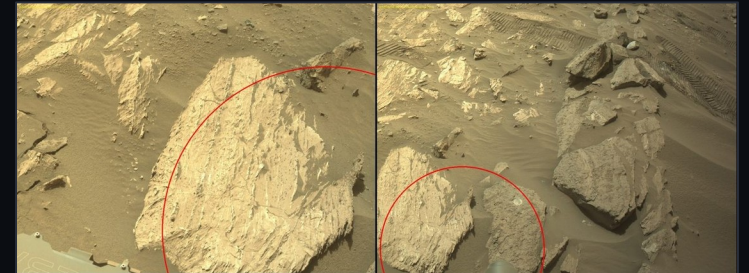
Every image ships with its camera model and the rover's position, so the scene is in metres from the first step. No structure-from-motion.



Fisheye Navcam frame, straightened with its JPL camera model



Depth from the two Navcam eyes, 42 cm apart
(red is near)



One spot (red circle) located in photos from other stops, from rover telemetry alone

0.1 px stereo matching error after undistorting, down from about 1 px

312,136 3D points from stereo, used as the starting positions

2 px worst misalignment between the three stops after registration (our limit: 3)

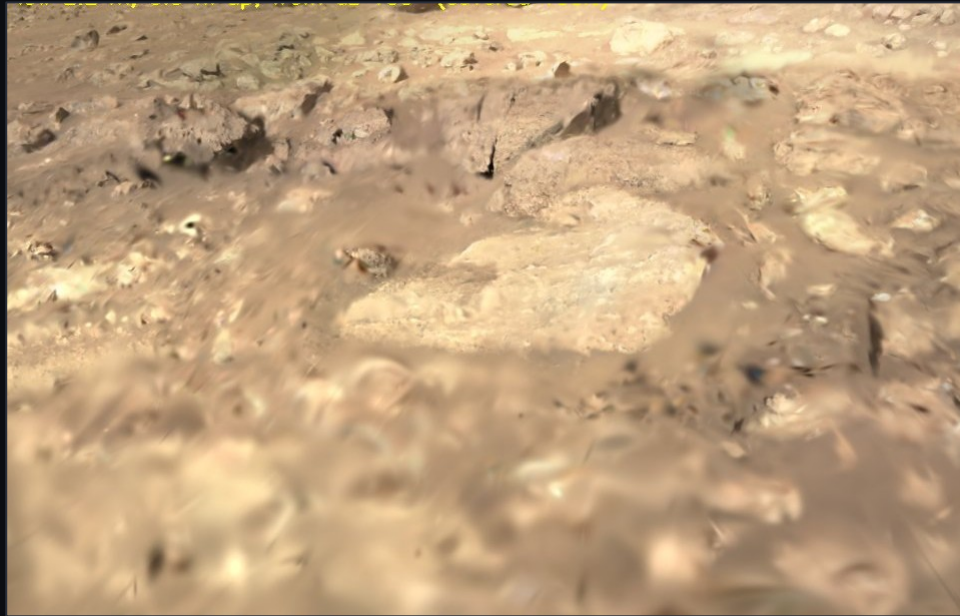
STEP 3

Train, then judge by looking

gsplat with the MCMC strategy on a Colab A100. The usual score barely moved between good and bad runs, so every run was rendered from new viewpoints.



First run: fine from the training cameras, spikes from anywhere else



Final run, same viewpoint

30,000

steps

25 min

on one A100

2.5 M

Gaussians

Fix 1 a depth-loss bug that produced NaNs

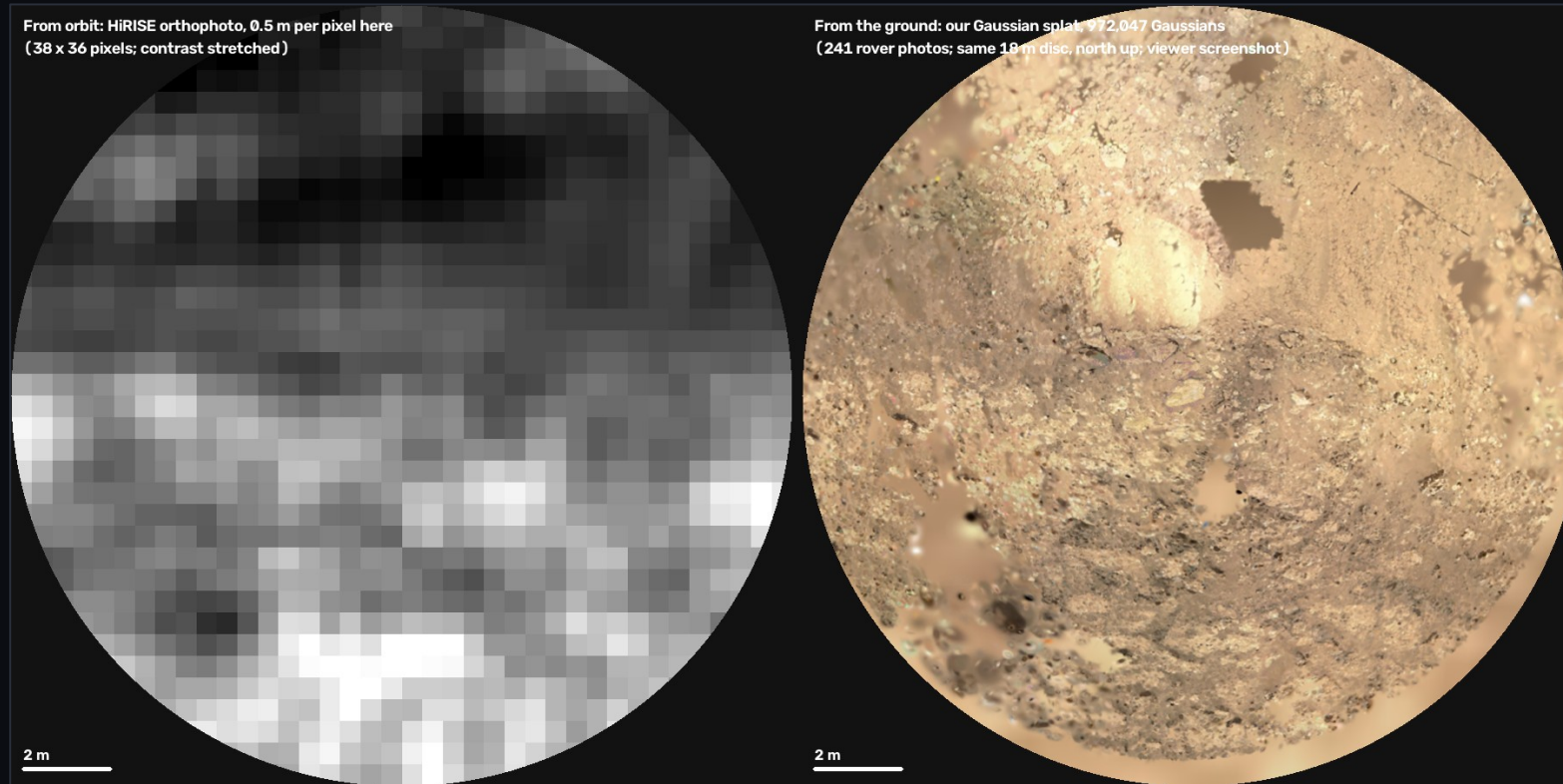
Fix 2 two regularisers that removed Gaussians seen by only a few photos

Fix 3 a penalty on needle-shaped Gaussians, which were the spikes

STEP 4

Put it on the map, and make it data

Cropped to the 9 m that the photos cover well, placed on NASA's orbital terrain in the same metric frame, with a layer value on every Gaussian.



The same 18 m disc: 38 x 36 orbital pixels on the left, our splat on the right

972,047

Gaussians in a 16 MB file

270

orbital height samples in the same disc

Layers

terrain class, roughness, search cluster on every Gaussian

10 cm

surface map the rover drives on

Limits: soft from closer than about 2 m, and holes where no photo covers the ground.